Silas Osunba

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Research Interests

- Applied Machine Learning
- Agentic AI
- Large Language Models
- eXplainable and Trustworthy AI

Education

LAGOS STATE UNIVERSITY, LAGOS, NIGERIA Sept. 2020 – Dec. 2025
Bachelor of Science (B.Sc.) in Electronic and Computer Engineering (Class Rank: 1/118)
Final Cumulative GPA: 4.77/5.0 **Top 1%** ***First Class Honors***
Language of Instruction: English
Thesis Title: Advanced Intelligent Framework For Modeling Rain-Induced Attenuation At Ka-Band
Relevant Coursework and Grades: *Computer Programming and Computations (A), Neural Networks (A), Computer Assisted Applications (A), Algorithms and Data Structure (A), Digital Signal Processing (A), Economics for Engineers (A), Digital Image Processing (A), Software Engineering (A), Operations Research (A), Digital Electronics (A), Control Engineering (A), Digital Control (A)*

Research Experience

□ **UNDERGRADUATE THESIS RESEARCH, LAGOS STATE UNIVERSITY** Lagos, Nigeria
Undergraduate Student Researcher *Jan. 2025 – Dec. 2025*
Advisor: Prof. A.I.O. Yussuff
Title: Advanced Intelligent Framework for Modeling Rain-Induced Attenuation at Ka-Band

- *Investigated the limitations of traditional rain-attenuation models in temperate and tropical regions to address gaps affecting satellite link reliability at Ka-band of 26 GHz.*
- *Proposed and developed an intelligent machine learning or deep-learning framework for modeling rain-induced attenuation, and evaluated its performance against traditional models.*
- *Achieved 20% improvement in prediction accuracy compared to baseline approaches, and outlined practical implications for network designers to optimize fade margins and improve the reliability of satellite links.*

□ **ENGINEERING.FIT() RESEARCH LABORATORY** Lagos, Nigeria
Machine Learning Researcher *Jun. 2024 – Sept. 2025*

- *Addressed slow response and high computational costs in oil spill detection by designing a CNN-based deep learning solution optimized for edge devices.*
- *Fine-tuned pre-trained models on custom oil spill datasets and applied float-16 quantization to enhance inference speed and reduce model size by 70% while maintaining 97% detection accuracy,*
- *Achieved faster inference and improved detection accuracy and efficiency by 10% and learned to apply advanced model optimization and deployment strategies for real-time and resource-limited environmental monitoring.*

□ **LAGOS STATE UNIVERSITY - Doyen Emerging Lab**
Department of Electronic and Computer Engineering, Lagos State University Lagos, Nigeria
Research Assistant & ML Researcher *Nov. 2023 – Mar. 2025*
Advisor: Dr. A.A Ajasa & Dr. M.A Adedoyin

- *Addressed challenges in predicting heart disease from high-dimensional clinical data, including low-variance features and complex interactions among indicators such as resting blood pressure and ECG results.*
- *Improved predictive performance of ensemble machine learning models through feature selection informed by previous research, hyperparameter tuning, and application of regularization techniques to prevent the model from overfitting.*
- *Achieved a 25% improvement in model accuracy, significantly increasing the model's ability to reliably identify the presence or absence of heart disease from patients' data.*

Conference Presentation / Paper Publication / Poster Presentation {J = Journal, C = Conference}

[J.1] Bakare, A. O., Seriki, K. T., & **Osunba, S. M.** (2025). *"Predicting Proton Exchange Membrane Fuel Cell Performance through Advanced Machine Learning Techniques: A Comparative Analysis of Ensemble Techniques."* Scientia. Technology, Science and Society, 2(8), 73-86. DOI: 10.59324/stss.2025.2(8).07

[C.1] A. O. Bakare, K. T. Seriki, F. O. Emmanuel, **S. T. Osunba**, S. E. Ayoigbala, and K. J. Oshile. *"Using Convolutional Neural Network for the Detection of Offshore and Onshore Oil Spills."* Paper presented at the SPE Nigeria Annual International Conference and Exhibition, Lagos, Nigeria, August 2025. DOI: 10.2118/228732-MS.

[C.2] O. Kolade, V. A. Adeleke, S. D. Ojo, **S. Osunba**, O. Anifowose, and K. Abdul. *"Electric vehicles vs. biofuels: A Review of Decarbonization in Transportation."* Paper presented at the SPE Nigeria Annual International Conference and Exhibition, Lagos, Nigeria, August 2025. DOI: 10.2118/228753-MS.

Honours, Awards & Achievements

- **Blockchain funding for AquOS** – *My AquOS project secured approximately $70,000 in market-cap funding from the blockchain community. AquOS is an open-source, experimental autonomous aquarium derivative project. The aim of the project is to promote scalable, zero-manual-maintenance intelligent aquaculture solutions for commercial breeding and individual enthusiasts.*

- **Chancellor's & Dean's Award for the Engineering Valedictorian** – *Faculty of Engineering and Department of Electronic and Computer Engineering, Class of 2025, Lagos State University.*
  Awarded by: Lagos State University Management and Senate, August 2026
- **Award of Excellence** – *Awarded by the Nigerian Society of Engineers for being the Best Student in the Faculty of Engineering, Class of 2025, Lagos State University.*
  Awarded by: Nigerian Society of Engineers, August 2026
  Awarded by: Blockchain Community
- **Outstanding Service Award** – *Honored for exemplary leadership and service as President of the SPE LASU Student Chapter during the 2024/2025 administration.*
  Awarded by: Society of Petroleum Engineers International (SPEI), September 2025
- **Scholar of the Department Award** – *Awarded for ranking as the top student in the Department of Electronic and Computer Engineering in my final-year class.*
  Awarded by: Association of Electronic and Computer Engineering Students, Lagos State University Chapter, September 2025
- **SAP Scholarship** – *A fully-funded scholarship to learn Systems, Applications, and Products (SAP) in Data Processing. It is worth $400.*
  WYZE Group - a firm of Chartered Accountants, Tax Advisers, Investment Specialists and Technology Consultants, July 2025
- **First E&P Scholar** – *Awarded for ranking as the top student in the Faculty of Engineering. The scholarship is worth $350*
  Awarded by: First Exploration and Petroleum Development Company & Society of Petroleum Engineers (SPE International), Lagos Section 61, August 2024
- **Third Runner-Up, DATICAN 1.0 Competition** – *My project was ranked 3rd runner-up out of 24 teams that participated in the University of Chicago, Data Science and Medical Image Analysis competition. My team won $100*
  Awarded by: University of Chicago in collaboration with Lagos State University. June 2024
- **4X DataCamp Scholarship** – *Recognized as one of 300 students across African universities selected to be empowered to learn and hone skills in Data Science and Artificial Intelligence. Each of the scholarship costs $250*
  Awarded by: DataCamp, December 2023
- **Undergraduate Scholarship Award for Excellent Performance in Engineering** – *Awarded by the Society of Petroleum Engineers for demonstrating academic excellence and leadership in engineering. The scholarship is worth $300*
  Awarded by: Society of Petroleum Engineers (SPE International), Lagos Section, July 2023
- **Vice Chancellor's Letters of Commendation/Award of Scholarship** – *Recognized as one of the distinguished First-Class students in my Sophomore year, and as a top student in my department, receiving a merit-based scholarship for outstanding academic achievements (4.89 CGPA on a scale of 5.0) in the 2020/2021 academic session.*
  Awarded by: Lagos State University Management and Senate, September 2022

Industry Experience

CONDERE CORP LTD — England, United Kingdom
AI & Product Engineer — *Jun. 2026 – Present*

- *Designing agentic AI and human-in-the-loop systems for elimination of 80% of manual recruitment processes.*

MONIEPOINT INC. — Victoria Island, Lagos, NG
Software Engineering Fellow — *Feb. 2026 – May 2026*

- *Designed and developed an analytics endpoints that queried merchant activity data to generate business insights for the Data Analytics and Growth teams.*
- *Deployed a containerized system that delivered business insights 75% faster and at 25% lower cost than previously deployed endpoints.*

Other Academic/Coursework Projects Completed

- **QueryMate** — See live project and details
  A python package for conversing with large relational database. It was designed with security-first at mind.
- **Training of MiniGPT** — See live project and details
  Implemented Generative Pretrained Transformer (GPT) models (e.g., GPT-2, GPT-3) for experimental purposes to gain a deeper understanding of the underlying mechanisms of this technology.
- **Automated Printed Circuit Board (PCB) Defect Detection System using YOLO** — See live project and details
  Designed and trained a PCB defect detection system using computer vision techniques and a YOLO-based pre-trained object detection model to identify visual defects in PCB images.
- **Design and Simulation of a Solar Power System for Domestic Use in MATLAB Simulink environment**
  Designed and simulated a 500W solar power system for hostel use in MATLAB Simulink, the solar setup includes all essential components of a typical solar power system to operate fans, phones, and other small electronic devices.
- **Automated Detection of Diabetic Retinopathy** — See live project and details
  Designed, developed and implemented an automated tool to assist ophthalmologists in detecting diabetic retinopathy in patients. The architectural design was inspired by Deep Residual Learning and EfficientNet.
- **Four Wheel Drive Robot Car Controlled Via Bluetooth**
  Designed and developed a bluetooth controlled 4WD robot car using Arduino Uno R3, an HC-05 Bluetooth module, and an L293D motor.

Skills And Competences

Programming & Software: *Python, MATLAB, JavaScript, Java, SQL, PyTorch, Tensorflow, NLTK, Hugging Face, MLflow, CI/CD, GenAI Frameworks (CrewAI, Langchain, LlamaIndex), Git, GitHub, Docker, Proteus, PSpice, PostgreSQL, MySQL, MongoDB, Latex, Linux, macOS*

Laboratory: *Electronic Circuit Design & Simulation, Sensor Calibration, Oscilloscope & Multimeter Operation, Data Acquisition Systems, Power Electronics Experimentation.*

Research: *Design of Experiment, Data Collection & Analysis, Technical Documentation, Manuscript writing*

Conference Organized / Attended

Conference Organized

[1] Nexus 2.0 Energy Policy and Innovation Conference, Epe, Lagos, Nigeria, July, 2025. Organized the annual Nexus 2.0 student-led energy conference as committee president. The conference featured 10 dignitaries and energy leaders; including speakers, panelists, judges, and energy researchers, and had over 500 students and 100 volunteers in attendance. It was nominated and also recognized as the best conference on campus.

Conferences Attended

[2] Nigeria Annual International Conference and Exhibition (NAICE), Lagos, Nigeria, August, 2025.
[3] 1st LASU Faculty of Engineering International Conference (LASUFEIC), Epe, Lagos, Nigeria, May, 2025.
[4] Sub-Saharan Africa International Petroleum Exhibition and Conference (SAIPEC), Lagos, Nigeria, February, 2025.
[5] Sub-Saharan Africa International Petroleum Exhibition and Conference (SAIPEC), Lagos, Nigeria, February, 2024.

Teaching Experience

LAGOS STATE UNIVERSITY, LAGOS, NIGERIA

Volunteer Tutor *Aug. 2022 – Sept. 2025*

- *Successfully tutored over 250 engineering students in core engineering courses; Mathematics, Fundamental of Electrical Engineering, Programming, Strength of Materials, Applied Mechanics, Science of Materials, and Control Engineering, achieving a remarkable pass rate of 60% among students.*

Invited Talks

[1] **QueryMate**. Invited Project Talk, **Cursor**, Lagos, Nigeria, April 2026.
[2] **Using Convolutional Neural Network for the Detection of Offshore and Onshore Oil Spills**. Invited Speaker, SPE Nigeria Annual International Conference and Exhibition (NAICE), Lagos, Nigeria, August 2025.

Leadership And Volunteering Experience

□ **Society of Petroleum Engineers International (SPEI)**

- President, SPE Lagos State University Student Chapter *Nov. 2024 – Oct. 2025*
  Oversaw executive operations and delegated responsibilities across executives and committees, ensuring efficient execution of academic, professional, and outreach projects or programs.
- Vice President, SPE Lagos State University Student Chapter *Nov. 2023 – Oct. 2024*
  Worked closely with the President to achieve and sustain necessary membership levels through targeted onboarding efforts.
- Membership Chair, SPE Lagos State University Student Chapter *Aug. 2023 – Nov. 2023*
  Oversaw and managed the membership activities of the organization.

□ **Nigerian Institute of Electrical and Electronic Engineers (NIEEE), LASU Student Chapter**

- Departmental Coordinator *Nov. 2024 – Oct. 2025*
  Maintained effective communication with every stakeholder involved for the progress of the organization; the President and other executives, our valued student members, and Faculty.

□ **Inspire The Student Initiative**

- Co-Founder *Sept. 2024 – Oct. 2025*
  - *Organized two profound webinars reaching over 150 participants, focused on empowering students with post-graduation success strategies.*
  - *Provided open-access study resources for all engineering departments (from second year to fifth year), hosting over 750 study materials on google drive.*
  - *Co-led a team of 15 dedicated volunteers to advance UN Sustainable Development Goal 4 (Quality Education) through mentorship, accessible learning and academic empowerment initiatives.*

Professional Membership & Affiliations

- Institute of Electrical and Electronics Engineers (IEEE) Oct. 2024 - Present
- Society of Petroleum Engineers International (SPEI) Aug. 2022 - Present
- International Association of Engineers (IAENG) Nov 2024 - Present